Arabic Summarization with a Seq2Seq Attention Model

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1. Introduction

Text summarization is one of the key applications of Natural Language Processing (NLP). It seeks to provide an accurate summary or abstract representation of an entire document by retaining the vital pieces of information contained within the latter. There are two major approaches employed in this regard, known as extraction-based and abstraction-based summarization techniques. The former extracts significant sentences from the text, while the latter creates summaries by generating new phrases. This paper will be concerned with the latter approach for texts written in the Arabic language.

1.1 Objectives

By the end of this project, the team aimed to:

1. Understand and implement Seq2Seq architectures with encoder and decoder networks.
2. include attention mechanisms (Additive Attention) to improve abstraction quality.
3. Perform Arabic text preprocessing including normalization, tokenization, and tashkeel removal.
4. Train and evaluate a deep learning model on Arabic summarization datasets.
5. Apply ROUGE evaluation metrics to assess summary quality objectively.

2. Datasets

The project combined five Arabic summarization datasets to maximize training data diversity and coverage of both standard Arabic and Egyptian dialect text.

Dataset	Source	Approx. Size	Description
Kaggle Arabic	Kaggle	~7,000	Arabic news articles with summaries (Processed Text / summarizer columns)
Summset	CSV corpus	~22,000	Arabic text-summary pairs with text / summarizer columns
SumArabic 1.0	Mendeley / JSONL	~84,000 (train/val/test)	MSA news articles with headlines as summaries; three-way split provided
Egyptian Arabic	Hugging Face / Parquet	~varies	Focused on Egyptian dialect; text / summarized text columns
AraSum (AbsSum)	GitHub / TSV	~49,600	Tab-separated monolingual Arabic corpus with article and summary pairs

2.1 Data Loading and Merging

Each dataset was loaded using a dedicated loader function that standardized column names to text (article body) and target (summary). Loaders handled encoding issues using encoding_errors='replace':

- SumArabic: loaded line-by-line from JSONL files, parsing text and headline fields.
- AraSum: loaded as TSV with quote-ignoring (csv.QUOTE_NONE) to handle malformed rows.
- Egyptian Arabic: loaded directly from Parquet format, which handles encoding natively.

After loading all five sources, the datasets were concatenated and deduplicated on the (article, summary) pair. The final merged dataset contained 68,984 unique article-summary pairs ready for training.

3. Text Preprocessing

Arabic text requires specialized preprocessing that differs significantly from English NLP pipelines. A clean_arabic_text() function was implemented to normalize all text before tokenization.

3.1 **Preprocessing Pipeline**

1. URL and HTML removal: stripped hyperlinks and HTML tags using regex.
2. Bidirectional Unicode control character removal: removed invisible formatting characters (U+200F, U+202A, etc.) common in web-scraped Arabic text.
3. Alef normalization: mapped all Alef variants (إأآا) to the base form (ا) to reduce vocabulary size.
4. Ta marbuta normalization: mapped ة to ه for consistency.
5. Ya normalization: mapped final Ya (ى) to standard Ya (ي).
6. Tashkeel (diacritic) removal: stripped harakat and other diacritical marks (Unicode range U+064B–U+065F, U+0670).
7. Punctuation removal: removed all ASCII punctuation characters.

After cleaning, rows where either the article or summary became shorter than 10 characters were discarded to ensure training quality.

3.2 Tokenization

Rather than training a vocabulary from scratch, the project used the pre-trained AraBERT tokenizer (aubmindlab/bert-base-arabertv2), which provides a vocabulary optimized for Arabic text using WordPiece subword tokenization. Key parameters:

- Maximum article length: 200 tokens (truncated/padded)
- Maximum summary length: 60 tokens (truncated/padded)
- Vocabulary size: estimated 64,000 tokens (inherited from AraBERT)

For training the decoder, target sequences were shifted: the decoder input is tokens 0 to N-2, and the target output is tokens 1 to N-1 (teacher forcing setup).

3.3 **Data Splits**

The preprocessed data was split using scikit-learn's train_test_split with a fixed random seed (42):

- Training set: 80% (~55,187 samples)
- Validation set: 10% (~6,899 samples)
- Test set: 10% (~6,898 samples)

4. Model Architecture

The system implements a bidirectional seq2seq model with additive attention using TensorFlow and Keras. Pre-trained AraBERT embeddings are used as frozen initializers, significantly reducing trainable parameters while leveraging rich Arabic word representations.

4.1 Encoder

5. Input: token IDs of shape (batch, 200)
6. Embedding layer: frozen AraBERT embeddings (768-dim), with mask_zero=True to handle padding
7. Bidirectional LSTM: 128 units in each direction (256 total). Returns full sequences and final states.
8. State concatenation: forward and backward hidden states (h, c) are concatenated to form the initial decoder state.

4.2 **Decoder**

Input: shifted target token IDs of shape (batch, 59)

Embedding layer: same frozen AraBERT embeddings (shared with encoder)

LSTM: 256 units, initialized with concatenated encoder final states. Returns full output sequences.

4.3 **Attention**

Additive (Bahdanau) Attention is applied between decoder outputs and encoder outputs. The attention context vector is computed as a weighted sum of encoder outputs, where weights reflect how relevant each encoder time step is to the current decoder time step. The context vector is concatenated with the decoder LSTM output before the final dense layer.

4.4 **Output**

A TimeDistributed Dense layer with softmax activation over the full vocabulary produces a probability distribution per decoder time step. The layer explicitly uses float32 dtype for numerical stability under mixed-precision training.

5. Training

Hyperparameter	Value
Batch Size	16 (kept small for GPU memory safety)
Max Epochs	8
Optimizer	Adam with default learning rate
Early Stopping Patience	4 epochs on validation loss
LR Reduction Factor	0.5 (patience: 2 epochs, min lr: 1e-6)
Model Checkpoint	Best validation loss saved to best_seq2seq_model.keras
History Logging	CSV logger saved epoch metrics to training_history.csv

6. Evaluation

6.1 **ROUGE Metrics**

ROUGE is the standard metric for text summarization. Three variants were computed:

- ROUGE-1: unigram overlap between generated and reference summary.
- ROUGE-2: bigram overlap, measuring fluency and coherence.
- ROUGE-L: longest common subsequence, capturing sentence-level structure.

The F-measure was used as the primary score. Evaluation was performed on a random sample of 20 test examples using the rouge_score library.

A second refined evaluation script additionally applied the Arabic text cleaning function to both generated and reference summaries before scoring, ensuring fair comparison by eliminating tokenization artefacts (such as ## WordPiece subword prefixes) from the comparison.

Our codes rouge evaluation gave us 0, due to the exact matching that rouge follows between reference and generated outputs. Our generated output could understand what the article was on, but could generate the words that exactly match the context.

[Sample 1]

Article: لحق رودا كيركراده بفيينورد روتردام الى المباراة النهائية لمسابقة كأس هولندا في كرة القدم بفوزه على مضيفه هيراكليس الميلو بركلات الترجيح 5-3 بعد انتهاء الوقتين الاصلي والاضافي بالتعادل 2-2 ...

Reference: رودا إلى نهائي كأس هولندا

Generated: كاس الاتحادي في دوري ابطال اوروبا

[Sample 2]

Article: اعلن نبيل بسباس رئيس الغرفه النقابيه لتجار المواد الاوليه والاحذيه والجلود خلال اجتماع عام مشترك عقد اليوم الاربعاء لتدارس الوضعيه المترديه ال اليها قطاع الجلود والاحذيه تراجع الانتاجيه ارباعها وقال ب...

Reference: كما يشكو القطاع من نقص في اليد العاملة وندرة في اليد العاملة المختصة.

Generated: وياتي هذا الاجتماع القضائي الاجتماعيه في قطاعات الدوله العموميه ، وفق بلاغ صادر عن وزاره الدفاع الوطني للاستثمار في التعليم الحالي

[Sample 3]

Article: اجلت الدائره الجنائيه بالمحكمه الابتدائيه بقرمباليه اليوم النظر قضيه حرق منزل شقيقه الرئيس الاسبق نعيمه بن علي بحمام بنت الجديدي الى جلسه يوم افريل القادم استجابه لطلب دفاع المتهمين يذكر شخصا صادره حق...

Reference: يذكر أن 18 شخصا صادرة في حقهم أحكام غيابية تراوحت بين عامين و 16 عاما اعترضوا على هذه الاحكام وهم يمثلون في القضية بحالة سراح.

Generated: تمكنت السلطات التونسيه للمرطه بالقطب الجزائري صباح اليوم الثلاثاء 27 فيفري 2017 ، اليوم الاربعاء عطله تسفير الشباب التونسي اليوم الجمعه

6.2 **Qualitative Evaluation**

A sanity check was performed by generating summaries for three randomly sampled test articles. Each output was compared side-by-side:

- The article text (first 200 characters)
- The reference (gold) summary
- The model-generated summary

7.Libraries

- TensorFlow / Keras: model definition, training, and inference
- Scikit Learn: train/validation/test splitting
- rouge_score: official ROUGE metric calculation
- pandas / NumPy: data loading, manipulation, and array operations
- Transformers

9. Discussion

Frozen pre-trained embeddings: Using AraBERT embeddings without fine-tuning was a deliberate choice to reduce GPU memory and training time. The 768-dimensional embeddings provide rich Arabic morphological and semantic information that a smaller model trained from scratch cannot replicate.

Bidirectional encoder: A bidirectional LSTM captures context from both directions of the input, which is especially important for Arabic where morphological agreement spans long distances.

Additive Attention: attention was chosen over Luong (multiplicative) attention because it tends to perform better in lower-resource training scenarios and is natively supported in Keras via AdditiveAttention.

Dataset merging strategy: combining five different datasets improves domain coverage and reduces overfitting to any single source's style. Deduplication ensures the model does not see the same pair multiple times.

10. Limitations and Challenges

The ROUGE evaluation sample size of 20 examples is small; a larger evaluation set would give more statistically reliable scores.

The maximum article length of 200 tokens may truncate long news articles, causing information loss.

Inference uses greedy decoding with temperature; beam search would likely produce higher ROUGE scores but was not implemented.

The model does not fully handle dialect diversity; while the Egyptian Arabic dataset is included, training data is majorly standard arabic.

Cells 3 and 4 are duplicates of the qualitative check, reflecting iterative development in the notebook.

11.Conclusion

The abstractive Arabic text summarization pipeline in this project was successfully created through a Bidirectional LSTM Seq2Seq with Additive Attention model using AraBERT pre-trained embeddings. All parts of the task have been taken care of such as multi-dataset loading and concatenation, specialized Arabic text preprocessing, tokenization using a pre-trained Arabic vocabulary model, model training with early stopping and learning rate scheduling, generation using greedy decoding with temperature scaling, and evaluation using ROUGE metrics.

The employment of pre-trained AraBERT embeddings as frozen initializers for the model was an important decision made in this experiment, taking advantage of semantic representations in Arabic without having to fine-tune the full BERT architecture, which could require more computations. The almost 69,000 examples dataset ensures broad representation of both standard arabic and Egyptian Arabic text.